

Abstract

Abstract I

The Cognitive Integrity Framework (CIF) [?] establishes formal guarantees for the safe operation of multiagent AI systems through five canonical defense mechanisms: Cognitive Firewalls, Belief Sandboxing, Behavioral Invariants, Drift Detection, and Byzantine Consensus. Paper 2 of this series [?] demonstrated 94–100% detection at the parametric ceiling across 950 attack scenarios and four production multiagent architectures, and Paper 3 [?] translated these results into accessible deployment guidance, incident-response playbooks, and operator risk frameworks. However, the practical applicability of these mechanisms across diverse operational domains has not been systematically evaluated. This paper—Part 4 of the *Cognitive Security for Multiagent Operators* series—addresses this gap.

Abstract II

We present the **CIF-AD-OODA integration model**, combining CIF's defense mechanisms with Axiomatic Design (AD) theory [?] and Boyd's OODA (Observe-Orient-Decide-Act) Loop [?] to analyze **Goal Hijacking**: an advanced form of indirect prompt injection that constitutes a *teleological attack* on autonomous agency. Unlike conventional exploits targeting data confidentiality, Goal Hijacking corrupts the OODA Loop's **Orientation** phase, overriding an agent's Functional Requirements (FRs) while preserving the appearance of rational behavior. We model these attacks as “fast OODA transients” that introduce off-diagonal coupling in the agent's Design Matrix, violating Suh's Independence Axiom.

Abstract III

Using a standardized five-step domain analysis template, we conduct a rigorous analysis across ten critical domains: (1) Rare Earth mining, (2) Nation-state alliances, (3) Cyber-security, (4) Drone warfare, (5) Supply chains, (6) Biowarfare, (7) Food security, (8) Trade wars, (9) Infrastructure, and (10) Information ecosystems. Cross-domain synthesis reveals three universal attack patterns—FR Polarity Inversion, Constraint Relaxation, and Context Boundary Violation—and confirms that all five CIF mechanisms provide adequate coverage across the domain portfolio. Three novel defense extensions emerge: *verification channel separation* (Biowarfare), *active perturbation probing* (Trade Wars), and *physics-informed invariants* (Infrastructure). Retrospective analysis of six documented AI agent security incidents (2024–2025)—including production database destruction, remote code execution via prompt injection, and \$3.2M procurement fraud—validates the attack pattern taxonomy and confirms CIF defense applicability. These findings are corroborated by the OWASP Top 10 for Agentic Applications (2025) [?], which designates Agent Goal Hijack as the primary risk for deployed